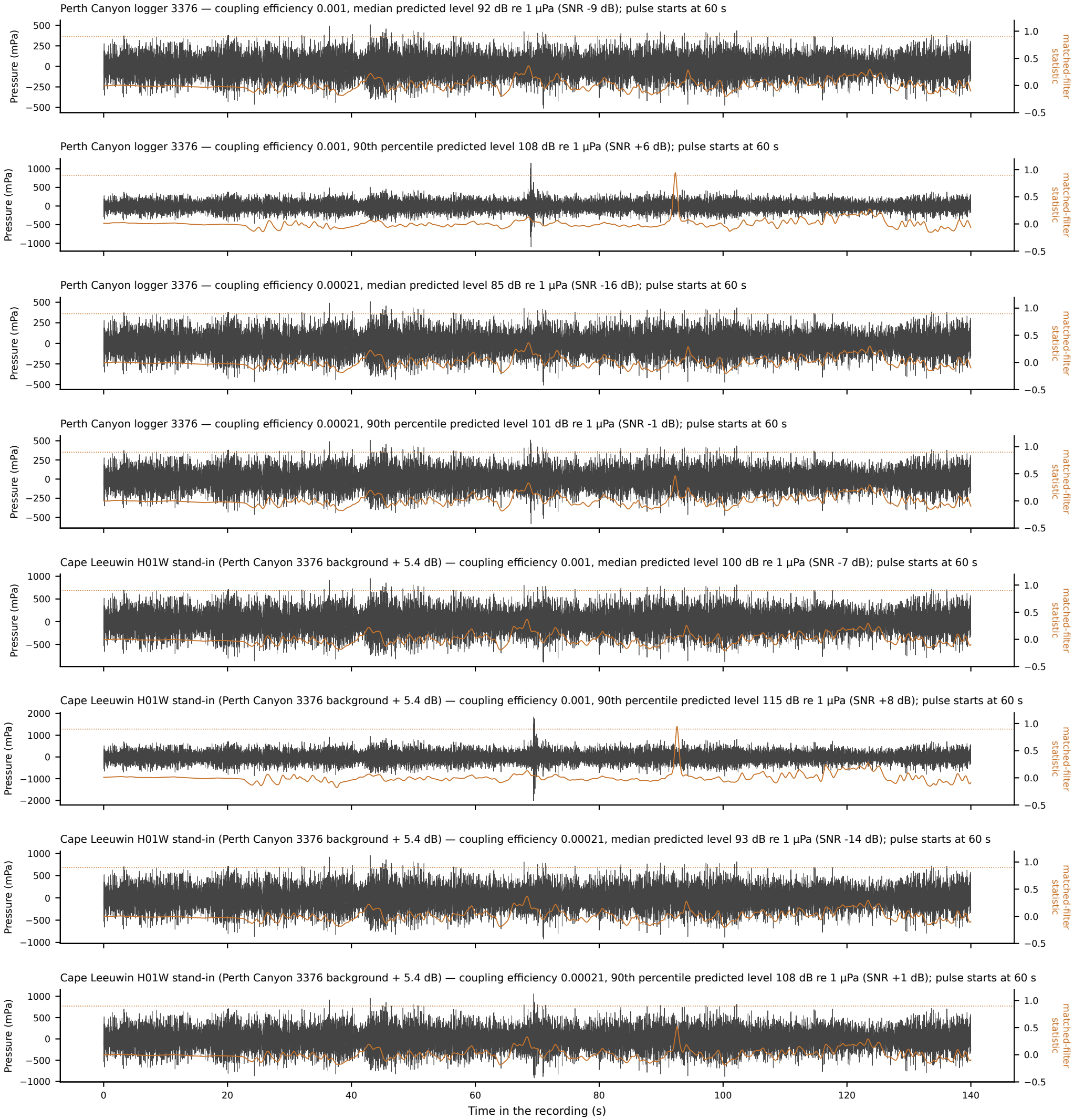
Illustrative only: a real background recording with a simulated impact pulse added at the predicted level, and the matched-filter statistic



Plain-language note: Each panel shows 140 s of a real recording from the Perth Canyon logger (grey, 5–40 Hz), with a simulated impact arrival added at 60 s at the level predicted for the median or the 90th-percentile impact under two values of the coupling efficiency. The orange trace is the matched-filter statistic used in the weak-signal trial; the dotted line is its false-alarm threshold on real background. For Cape Leeuwin the same recording is used as a stand-in, raised to that station's noise level. This is an illustration, not a detection.

Technical note: 3376 recording MH370/3376_PerthCanyon/531A5E9E.DAT (2014-03-08 00:05:02 logger clock, role B), calibrated with the Curtin gain file and -196/-197.8 dB sensitivity, 5-40 Hz zero-phase Butterworth; pulse = the weak-signal track's KRAKEN dispersed template (templates_raw.npz h_2376.1 / h_2200.0, resampled to 500 Hz) scaled so its peak 1 s mean-square equals the predicted level (D0', p = 0.5, 'level_q50'/'level_q90' from detectable.csv, +4.8 dB at H01W); statistic = normalised cross-correlation of the 0.5 s squared envelope with the template envelope (M1b-type); 0.897 = alpha 0.01 null threshold at 3376 from the weak-signal note. H01W stand-in = 3376 background + Blackman offset 5.44 dB. PREDICTIVE, NOT CALIBRATED; caveats C1 (propagation slope misfit, now a source-term caveat after the EW0112 result) and C2 (coupling efficiency not calibrated; F-35A upper anchor 2.1e-4).